

## FINCASTLE AGDM, AB, Canada

WMO: 715350

Lat: 49.8023N

Lon: 112.0459W

Elev: 804

StdP: 92.03

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -27.6      | -24.5 | -31.0                         | 0.2 | -27.4 | -27.7 | 0.3 | -24.4 | 13.7                  | 4.4  | 12.4 | 1.3  | 2.9                   | 280  | 0.669 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.3                   | 31.8            | 17.1 | 29.9 | 16.8 | 28.1 | 16.1 | 19.1                | 27.6 | 18.1 | 26.8 | 17.1 | 25.6 | 3.8                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.1                            | 12.7 | 22.6 | 15.0 | 11.7 | 21.5 | 13.9 | 10.9 | 20.3 | 57.8          | 27.6 | 54.4 | 26.6 | 51.3 | 25.7 | 24.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 12.4          | 10.9 | 9.4 | -32.9                      | 35.5 | 3.1                | 1.7 | -35.1                                              | 36.7 | -36.8      | 37.7 | -38.6      | 38.7 | -40.8      | 40.0 |    |    |
| (5)               |      |     | -33.0                      | 21.5 | 3.1                | 1.3 | -35.2                                              | 22.5 | -37.1      | 23.3 | -38.8      | 24.0 | -41.1      | 25.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 6.2    | -5.4  | -6.3  | 0.2   | 6.1   | 11.9  | 15.9  | 19.3  | 18.1  | 13.3  | 6.7   | -0.2  | -6.3  |  |  |
|                                                                        | DBStd       | 11.06  | 8.80  | 8.82  | 7.46  | 5.01  | 4.35  | 3.15  | 2.78  | 3.25  | 4.52  | 5.10  | 7.47  | 8.52  |  |  |
|                                                                        | HDD10.0     | 2362   | 479   | 456   | 304   | 132   | 29    | 1     | 0     | 1     | 19    | 126   | 309   | 507   |  |  |
|                                                                        | HDD18.3     | 4552   | 737   | 689   | 561   | 367   | 203   | 86    | 22    | 45    | 159   | 362   | 556   | 765   |  |  |
|                                                                        | CDD10.0     | 964    | 0     | 0     | 2     | 15    | 89    | 178   | 288   | 250   | 118   | 22    | 3     | 0     |  |  |
|                                                                        | CDD18.3     | 112    | 0     | 0     | 0     | 0     | 3     | 13    | 52    | 37    | 8     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 2071   | 0     | 0     | 0     | 12    | 122   | 237   | 819   | 652   | 221   | 7     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 706    | 0     | 0     | 0     | 2     | 26    | 60    | 303   | 243   | 71    | 1     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 4.4    | 5.0   | 4.3   | 4.7   | 5.0   | 4.5   | 4.1   | 3.4   | 3.3   | 3.8   | 4.5   | 4.8   | 4.8   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 377    | 22    | 14    | 23    | 35    | 45    | 61    | 34    | 43    | 40    | 16    | 17    | 21    |  |  |
|                                                                        | PrecMax     | 649    | 68    | 29    | 45    | 147   | 135   | 181   | 120   | 121   | 133   | 47    | 44    | 57    |  |  |
|                                                                        | PrecMin     | 244    | 2     | 0     | 0     | 0     | 8     | 7     | 9     | 6     | 5     | 0     | 2     | 4     |  |  |
|                                                                        | PrecStd     | 103    | 17    | 8     | 12    | 29    | 31    | 41    | 25    | 31    | 33    | 12    | 11    | 14    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 13.0  | 14.0  | 18.2  | 25.1  | 29.6  | 31.6  | 34.4  | 33.8  | 31.7  | 24.2  | 18.1  | 11.7  |  |  |
|                                                                        |             | MCWB   | 6.9   | 6.1   | 8.7   | 11.4  | 14.6  | 17.9  | 17.5  | 17.3  | 15.4  | 12.2  | 9.8   | 4.8   |  |  |
|                                                                        | 2%          | DB     | 9.5   | 9.7   | 15.4  | 21.0  | 26.7  | 28.2  | 32.0  | 31.6  | 28.8  | 20.5  | 13.9  | 9.0   |  |  |
|                                                                        |             | MCWB   | 4.7   | 4.1   | 7.4   | 9.8   | 13.7  | 16.3  | 18.2  | 16.6  | 14.7  | 10.8  | 7.2   | 4.1   |  |  |
|                                                                        | 5%          | DB     | 7.3   | 7.3   | 13.1  | 18.0  | 24.0  | 26.1  | 30.0  | 29.4  | 25.9  | 17.6  | 11.5  | 6.8   |  |  |
|                                                                        |             | MCWB   | 3.1   | 2.8   | 5.9   | 8.5   | 12.4  | 15.2  | 17.8  | 16.1  | 13.9  | 9.6   | 6.1   | 2.9   |  |  |
|                                                                        | 10%         | DB     | 5.4   | 5.2   | 10.4  | 15.2  | 21.3  | 23.9  | 28.0  | 27.2  | 22.7  | 15.1  | 9.2   | 5.0   |  |  |
|                                                                        |             | MCWB   | 2.1   | 1.6   | 4.6   | 7.1   | 11.2  | 14.4  | 17.1  | 15.7  | 12.9  | 8.2   | 4.4   | 1.4   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 7.5   | 6.6   | 9.6   | 12.0  | 15.5  | 19.4  | 21.3  | 19.3  | 17.1  | 13.4  | 10.3  | 6.1   |  |  |
|                                                                        |             | MCDB   | 12.4  | 12.6  | 16.6  | 23.5  | 27.3  | 27.8  | 29.4  | 28.7  | 26.4  | 21.7  | 17.3  | 10.2  |  |  |
|                                                                        | 2%          | WB     | 4.9   | 4.5   | 7.8   | 10.3  | 14.0  | 17.7  | 19.7  | 18.1  | 15.5  | 11.7  | 7.7   | 4.2   |  |  |
|                                                                        |             | MCDB   | 9.5   | 9.4   | 14.5  | 20.1  | 25.0  | 25.6  | 27.9  | 27.6  | 26.0  | 19.2  | 13.0  | 8.0   |  |  |
|                                                                        | 5%          | WB     | 3.4   | 3.0   | 6.4   | 8.8   | 12.9  | 16.4  | 18.7  | 17.2  | 14.3  | 10.0  | 6.1   | 2.8   |  |  |
|                                                                        |             | MCDB   | 6.8   | 7.0   | 12.6  | 16.9  | 22.4  | 24.3  | 27.1  | 26.5  | 24.5  | 16.7  | 10.8  | 6.2   |  |  |
|                                                                        | 10%         | WB     | 2.2   | 1.6   | 4.7   | 7.5   | 11.9  | 15.1  | 17.8  | 16.3  | 13.1  | 8.7   | 4.6   | 1.5   |  |  |
|                                                                        |             | MCDB   | 5.2   | 4.9   | 10.0  | 14.3  | 20.2  | 22.2  | 26.3  | 25.1  | 22.1  | 14.5  | 8.6   | 4.6   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 10.5  | 11.3  | 11.7  | 12.7  | 13.2  | 12.6  | 15.3  | 15.4  | 14.3  | 12.4  | 11.0  | 10.2  |  |  |
|                                                                        |             | MCDBR  | 10.7  | 13.0  | 15.6  | 18.5  | 18.2  | 16.9  | 19.3  | 20.4  | 20.4  | 16.5  | 13.6  | 10.9  |  |  |
|                                                                        |             | MCWBR  | 7.2   | 8.3   | 8.7   | 9.1   | 7.6   | 7.2   | 8.0   | 7.9   | 8.4   | 8.5   | 8.1   | 7.2   |  |  |
|                                                                        | 5% WB       | MCDBR  | 10.5  | 12.9  | 15.1  | 17.2  | 16.3  | 15.6  | 17.1  | 17.6  | 19.0  | 15.5  | 12.9  | 10.8  |  |  |
|                                                                        |             | MCWBR  | 7.3   | 8.7   | 8.9   | 8.8   | 7.1   | 7.2   | 7.9   | 7.3   | 8.0   | 8.2   | 8.1   | 7.5   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.262 | 0.269 | 0.292 | 0.333 | 0.350 | 0.345 | 0.347 | 0.366 | 0.325 | 0.290 | 0.263 | 0.250 |  |  |
|                                                                        | taud        |        | 2.450 | 2.481 | 2.471 | 2.355 | 2.358 | 2.420 | 2.428 | 2.348 | 2.438 | 2.517 | 2.497 | 2.437 |  |  |
|                                                                        | Ebn at Noon |        | 811   | 896   | 927   | 912   | 907   | 912   | 905   | 867   | 874   | 855   | 804   | 774   |  |  |
|                                                                        | Edh at Noon |        | 59    | 74    | 89    | 112   | 117   | 111   | 108   | 112   | 92    | 70    | 56    | 52    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.25  | 2.24  | 3.54  | 4.88  | 5.81  | 6.41  | 7.01  | 5.60  | 3.98  | 2.48  | 1.39  | 0.97  |  |  |
|                                                                        | RadStd      |        | 0.11  | 0.18  | 0.27  | 0.33  | 0.45  | 0.42  | 0.36  | 0.35  | 0.33  | 0.25  | 0.10  | 0.09  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (65 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page